

# **Traffic Accident Severity and Emergency Response Prediction Using Machine Learning**

## **1. Introduction**

Traffic accidents remain a major global public safety issue, causing injuries, fatalities, and significant economic losses. Rapid and accurate estimation of accident severity is crucial for effective emergency response planning and traffic management.

This project aims to develop a machine learning system capable of predicting accident severity levels based on spatial, temporal, and environmental features. The ultimate goal is to support intelligent emergency decision-making and resource allocation.

## **2. Project Objective**

The objective of this project is to:

- Predict accident severity (Levels 1–4)
- Estimate emergency response urgency
- Identify spatial and temporal risk patterns
- Deploy a real-time web-based prediction system

### **3. Real-World Problems Addressed**

This project addresses several practical challenges:

1. Delayed emergency response due to lack of early severity estimation.
2. Inefficient distribution of emergency resources.
3. Limited awareness of hazardous environmental driving conditions.
4. Difficulty identifying accident-prone geographic regions.

### **4. Dataset Description and Feature Justification**

The project utilizes the US Accidents dataset (2016–2023), containing over 1.4 million records and 46 features.

However, not all features contribute meaningfully to severity prediction. Therefore, a focused feature selection strategy was applied.

Selected features:

- Start\_Lat
- Start\_Lng
- Temperature(F)
- Visibility(mi)

- Sunrise\_Sunset
- Hour
- Month
- DayOfWeek
- Weather\_Condition
- Severity (Target)

### **Justification of Feature Selection**

Geographic Coordinates:

Feature importance analysis later confirmed that location contributes over 55% of predictive power. Accident severity varies significantly across regions.

Weather Features:

Temperature and visibility directly affect road safety and driver behavior.

Temporal Features:

Traffic congestion and driver fatigue patterns vary by hour and day.

Unstructured text features such as Description were excluded to avoid increasing dimensionality and model complexity without structured transformation.

## **5. Data Cleaning and Exploratory Analysis**

Data cleaning was not limited to handling missing values. It included structural analysis and behavioral pattern discovery.

### **5.1 Handling Missing Values**

Rows with missing values in critical predictive features were removed.

Rationale:

- Severity prediction relies heavily on environmental conditions.
- Imputation could introduce artificial correlations.
- Large dataset size allowed safe removal without significant data loss.

### **5.2 Time Feature Extraction**

From Start\_Time, the following were extracted:

- Hour
- Month
- DayOfWeek

This allowed capturing temporal behavior patterns.

### **5.3 Peak Hour Analysis (Morning and Evening Peaks)**

Exploratory data analysis revealed two significant traffic peaks:

- Morning peak: 7 AM – 9 AM
- Evening peak: 4 PM – 7 PM

These peaks were clearly visualized in hourly distribution plots.

This finding is critical because:

- Higher traffic density increases accident impact.
- Severity tends to rise during rush hours.

Thus, Hour became a highly informative feature in the model.

### **5.4 Categorical Encoding**

Weather\_Condition was transformed using One-Hot Encoding.

Why?

Machine learning algorithms require numerical input. One-hot encoding preserves categorical information without imposing artificial ordinal relationships.

### **5.5 Class Imbalance Analysis**

The dataset exhibited severe class imbalance:

- Severity 2  $\approx$  79% of data
- Severity 1 and 4 are rare

This imbalance significantly influenced model behavior.

## **6. Model Training and Performance**

Two models were trained:

- Decision Tree
- Random Forest

Split ratio: 80% training / 20% testing

Random seed: 42

### **6.1 Decision Tree Performance**

Accuracy: 80.61%

Strength:

- Easy interpretability

Weakness:

- Sensitive to imbalance
- Lower minority class recall

### **6.2 Random Forest Performance**

Accuracy: 81.56%

Advantages:

- Reduced overfitting

- Captured nonlinear relationships
- Provided feature importance

Random Forest showed improved precision in minority classes but still struggled with recall for rare severity levels.

## **7. Failure in Predicting Rare Classes**

Both models struggled to predict rare classes (Severity 1 and 4).

This occurred because:

- Models were biased toward dominant class (Severity 2).
- The dataset imbalance affected recall performance.

This is evident in confusion matrices and F1-scores.

## **8. Practical Mitigation Through Intelligent Filtering (Hybrid Approach)**

To address this limitation in deployment, a practical rule-based enhancement was implemented inside `app.py`.

Additional logic layer:

If:

- $\text{Visibility} \leq 1.5 \text{ miles}$
- $\text{Temperature} \leq 20^{\circ}\text{F}$

Then:

Severity is automatically set to Level 4.

Additionally:

If accident occurs during peak hours and predicted severity is low, severity is elevated to Level 3.

This hybrid approach combines:

- Statistical machine learning
- Domain-driven expert rules

This significantly improves real-world reliability and safety responsiveness.

## **9. Feature Importance Findings**

Random Forest revealed:

1. Start\_Lng – 28.7%
2. Start\_Lat – 27.9%
3. Temperature – 14.7%
4. Hour – 9%

Geographic location dominates predictive influence.

This confirms that accident severity is spatially dependent.



## **10. Statistical Reliability**

Random Forest accuracy: 81.56%

Test samples: 1,470,825

95% Confidence Interval:

81.56%  $\pm$  0.07%

The extremely narrow interval confirms high statistical stability.

## **11. Deployment Architecture**

The trained model was saved using joblib and deployed using Flask.

Deployment steps:

1. Collect user input
2. Convert input to DataFrame
3. Add missing encoded columns
4. Reorder columns
5. Predict severity
6. Apply rule-based override
7. Return readable output

This ensures consistent feature alignment and reliable predictions.

## **12. Key Challenges**

- Severe class imbalance
- Minority class prediction weakness
- Managing encoded features during deployment
- Ensuring prediction consistency

## **13. Benefits of the Project**

### **General Benefits**

- Enhances emergency response planning
- Improves traffic risk awareness
- Supports intelligent transportation systems
- Assists urban safety planning

### **Analytical Benefits**

From Data Analysis:

- Identified spatial risk hotspots
- Discovered morning and evening accident peaks
- Measured environmental impact on severity

From Model Training:

- Demonstrated ensemble superiority

- Quantified feature importance
- Evaluated statistical reliability

From Deployment:

- Converted static model into real-time system
- Integrated hybrid intelligence logic

#### **14. Overall Contribution**

This project demonstrates the full pipeline of machine learning:

Data → Cleaning → Feature Engineering → Modeling →  
Evaluation → Deployment → Real-world adaptation.

It bridges theoretical machine learning concepts with practical emergency response applications.

#### **15. Conclusion**

The developed system successfully predicts traffic accident severity using structured environmental and spatial features.

Although class imbalance limits minority class recall, the integration of a hybrid rule-based enhancement improves real-world reliability.

The project provides both analytical insight and deployable intelligence, making it a strong foundation for smart transportation and emergency response systems.

Future work may include:

- SMOTE balancing
  - Deep learning approaches
  - Real-time traffic integration
  - Geographic generalization testing
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